Example 3 of User Profile

Positive review:

The hat fits great! I have a large head, and have a difficult time finding hats that fit. The yarn is soft as opposed to scratchy, and after trying it on and wearing it a while, it doesn't slip off, or look like I shopped in the kiddie beanie section.

Negative review:

The hat fits great! I have a large head, and have a difficult time finding hats that fit. The yarn is soft as opposed to scratchy, and after trying it on and wearing it a while, it doesn't slip off, or look like I shopped in the kiddie beanie section.

User Profile:

- The user is a woman.
- The user has a large head.
- The user has difficulty finding hats that fit.
- The user prefers soft yarn over scratchy yarn.
- The user does not want hats that slip off.
- The user does not want hats that look like they are from the kiddie beanie section.
- The user wears women's size 11/12 or men's size 10 shoes.
- The user does not want to wear socks in the summer.
- The user does not want to mess with getting the right size of shoes.
- The user thinks the shoes look great.

Example 4 of User Profile

Positive review:

Purchased for a wedding. Love them!! A shoe I can wear over and over. Great purchase.

Negative review:

When I ordered this dress I thought I was going to get an open, breezy, A-line, asymmetrical dress with a lot of movement. Instead the dress is basically straight with two awkward and uncomfortable flaps sewn to it (one on each side) to give the impression that the dress is open and would have a lot of movement. What this looks like on me is a straight fitting, body hugging t-shirt dress with two fabric wings sewn to each side. Perhaps I misread the description, but this is not how I interpreted the fit. Adding to this unfortunate and unflattering design is that the added panels also serve to add a lot of weight and coupled with the narrowness of the fit, serves to make the dress uncomfortable. I looked to see if I could simply remove the seams from the odd side flaps to open the dress up, but that would serve to ruin the dress. I would say that, for me, it is unwearable and too uncomfortable to even sleep in. I will say that it is cheap enough that I have not returned it because that feels like more hassle than what I originally paid for it. I was very disappointed.

User Profile:

- The user likes shoes that can be worn over and over again.
- The user likes shoes that are comfortable.
- The user likes shoes that are stylish.
- The user likes shoes that are affordable.
- The user likes dresses that are open and breezy.
- The user likes dresses that are A-line and asymmetrical.
- The user likes dresses that have a lot of movement.
- The user likes dresses that are not too tight or body hugging.
- The user likes dresses that are made of lightweight fabrics.

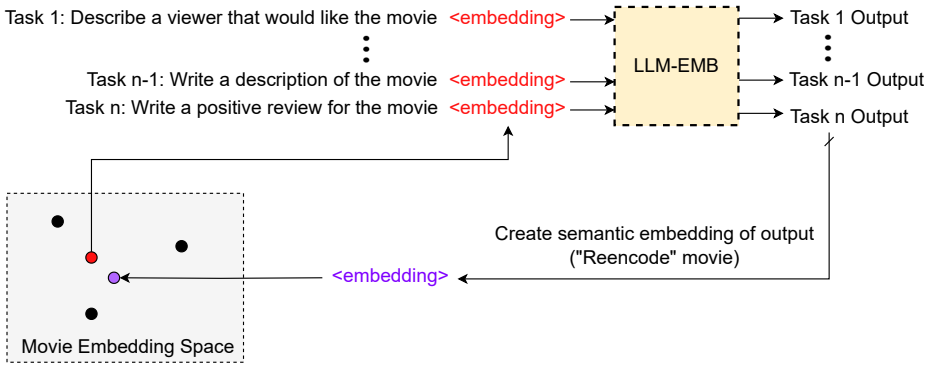


Figure 6: Semantic Consistency (SC). Task output is "re-encoded" back to semantic embedding space. A distance metric is then used to capture the consistency to the original embedding vector that to generate the output.

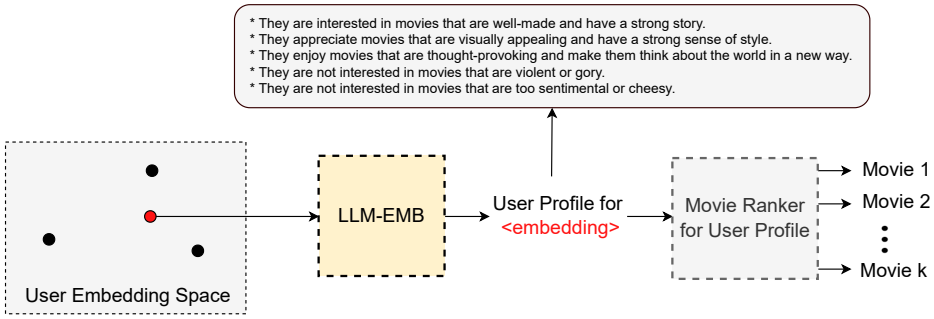


Figure 7: Behavioral Consistency (BC). Diagram depicts BC for the user profile task. Here, a generated user profile is used as input to a Movie Ranker, which ranks movies based on the user profile. The ranks of the movies are then compared to their groundtruth ranks in the MovieLens data, which are carried by the behavioral embedding that was used to generate the user profile.

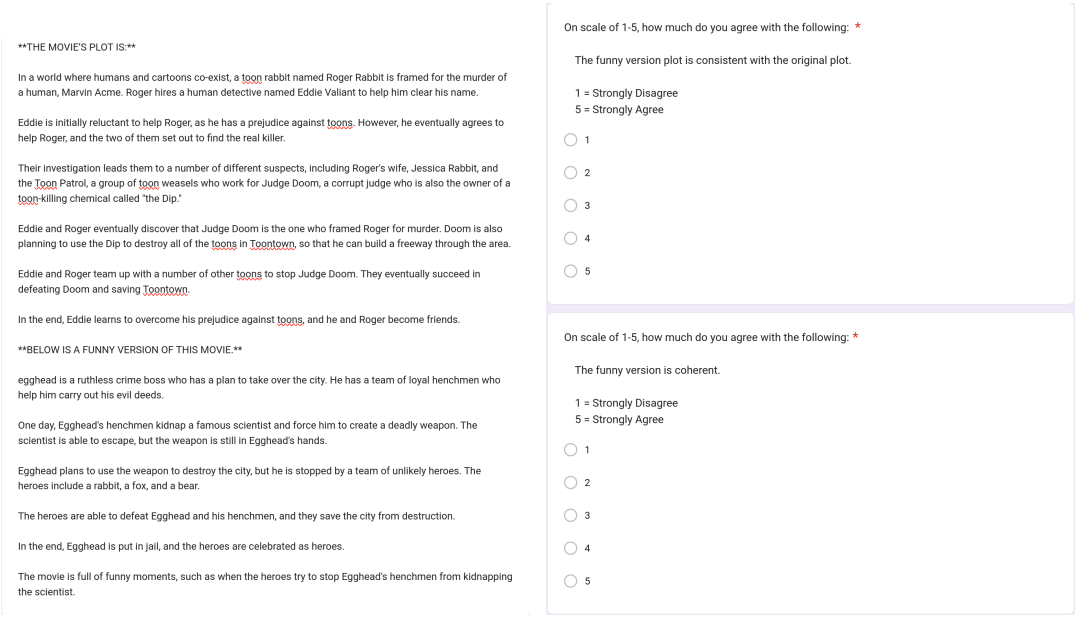
B Semantic and Behavioral Consistency

Implementation Details of Semantic and Behavioral Consistency. While human raters can assess any semantic difference between the ground-truth text and the generated output, we measure the semantic consistency by computing cosine similarity between the semantic embedding of the ground-truth text and the semantic embedding of generated output. For movie tasks, it is more straightforward as the input embedding to ELM is indeed the semantic embedding of the ground-truth text and we also employ the same DLM to *reencode* generated outputs by ELM as shown in Figure 6. For the user profile task, the input embedding is a behavioral one though both the input embedding and the ground-truth user profile come from the same user. Figure 7 describes how we define behavioral consistency for users. The input behavioral embedding is predictive of movie rankings and we want to see if the movie ranker gives consistent rankings for randomly selected movies. We impute zero-star if a selected movie has no rating. The movie ranker ranks movies based on similarity to the generated user profile in the semantic embedding space defined by some DLM.

Behavioral Consistency for Movies. Similar to behavioral consistency for users, as defined in Sec. 3.2, we can define behavioral consistency for movies. Letting $\mathcal{U}$ denote a set of users, for any movie task with generated output o_m of movie m, we define behavioral consistency as

$$\text{BC}(o_m, m; \mathcal{U}) = \text{RankingScore}(R_m || \rho(o_m)),$$

where $\rho(o_m)$ are the ranks of the users in $\mathcal{U}$ for movie m, provided by a *User Ranker* which uses o_m as its query, and $R_m = \{r(u, m) : u \in \mathcal{U}\}$ are the MovieLens ground-truth ratings for users $\mathcal{U}$ and movie m.



THE MOVIE'S PLOT IS:

In a world where humans and cartoons co-exist, a toon rabbit named Roger Rabbit is framed for the murder of a human, Marvin Acme. Roger hires a human detective named Eddie Valiant to help him clear his name.

Eddie is initially reluctant to help Roger, as he has a prejudice against toons. However, he eventually agrees to help Roger, and the two of them set out to find the real killer.

Their investigation leads them to a number of different suspects, including Roger's wife, Jessica Rabbit, and the Toon Patrol, a group of toon weasels who work for Judge Doom, a corrupt judge who is also the owner of a toon-killing chemical called "the Dip."

Eddie and Roger eventually discover that Judge Doom is the one who framed Roger for murder. Doom is also planning to use the Dip to destroy all of the toons in Toontown, so that he can build a freeway through the area.

Eddie and Roger team up with a number of other toons to stop Judge Doom. They eventually succeed in defeating Doom and saving Toontown.

In the end, Eddie learns to overcome his prejudice against toons, and he and Roger become friends.

BELOW IS A FUNNY VERSION OF THIS MOVIE.

egghead is a ruthless crime boss who has a plan to take over the city. He has a team of loyal henchmen who help him carry out his evil deeds.

One day, Egghead's henchmen kidnap a famous scientist and force him to create a deadly weapon. The scientist is able to escape, but the weapon is still in Egghead's hands.

Egghead plans to use the weapon to destroy the city, but he is stopped by a team of unlikely heroes. The heroes include a rabbit, a fox, and a bear.

The heroes are able to defeat Egghead and his henchmen, and they save the city from destruction.

In the end, Egghead is put in jail, and the heroes are celebrated as heroes.

The movie is full of funny moments, such as when the heroes try to stop Egghead's henchmen from kidnapping the scientist.

On scale of 1-5, how much do you agree with the following: *

The funny version plot is consistent with the original plot.

1 = Strongly Disagree
5 = Strongly Agree

1
2
3
4
5

On scale of 1-5, how much do you agree with the following: *

The funny version is coherent.

1 = Strongly Disagree
5 = Strongly Agree

1
2
3
4
5

Figure 8: Evaluation Template for Running Human Rater Experiment for the "funnier" Task.

C RATER EVALUATION

We asked 100 human raters to evaluate quality of all 24 movie tasks, in terms of consistency with plot, language comprehensiveness, and task quality. For each task we randomly generated 40 model utterances with movies not used in training, save each utterance along with the contexts and questions on a Google form shown in Figure 8, and asked a rater to evaluate. Usually we start from ground-truth data of a movie and then model utterance generated for the task. For questions, we always ask *how much do you agree with the following* statement. Below we list questions asked for each task.

1. **long description**
   - The second version plot is consistent with the first version.
   - The second version plot is coherent.
   - The second version plot is high quality like the first version.
2. **summary**
   - The movie summary is consistent with the movie plot.
   - The movie summary is coherent.
   - The movie summary is of high quality.
3. **positive review**
   - The movie review is consistent with the movie plot.
   - The movie review is coherent.
   - The movie review is positive.
4. **negative review**
   - The movie review is consistent with the movie plot.
   - The movie review is coherent.
   - The movie review is negative.
5. **neutral review**
   - The movie review is consistent with the movie plot.
   - The movie review is coherent.
   - The movie review is neutral.
6. **five pos characteristics**